

Lab - 3 : The logistic equation and its modifications

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In this lab, we analyzed different dynamical equations related to population growth and innovation spread using analytical and numerical methods. We explored the logistic growth equation, Euler's numerical integration, and modifications to the logistic model, comparing theoretical solutions with numerical approximations to evaluate accuracy.

I. MODEL

The human population $x(t)$ follows the logistic equation:

$$\dot{x} = ax - bx^2 \quad (1)$$

The limiting population is given by:

$$x_{\lim} = \frac{a}{b} \quad (2)$$

The analytical solution of the logistic equation is:

$$x(t) = \frac{x_{\lim}}{1 + \left(\frac{x_{\lim} - x_0}{x_0} \right) e^{-at}} \quad (3)$$

The logistic equation is modified as

$$\dot{x} = f(x) = rx[1 - (x/k)] - h \quad (4)$$

Where h is the “harvesting” rate.

The $x(t)$ by Euler's method is given by

$$x(n+1) = x(n) + f(x(n))\Delta t \quad (5)$$

The dynamical equation for the spread of agricultural innovations among farmers through personal communications is given by

$$\dot{x} = Cx(n-x) \quad (6)$$

The $x(t)$ for the spread of agricultural innovations among farmers through personal communications is given by

$$x = \frac{Ne^{CNt}}{(N-1) + e^{CNt}} \quad (7)$$

The dynamical equation for the spread of agricultural innovations among farmers through impersonal communications is given by

$$\dot{x} = (Cx + C')(n-x) \quad (8)$$

The $x(t)$ for the spread of agricultural innovations among farmers through impersonal communications is given by

$$x = \frac{NC'[1 - e^{-(CN+C')t}]}{C' + CNe^{-(CN+C')t}} \quad (9)$$

Defining $X = x/N$, $T = cNt$, $A = C/CN$ and $\dot{X} = dX/dT$,

Recast dynamical equation for the spread of agricultural innovations among farmers through personal communications is given by

$$\dot{X} = X(1-X) \quad (10)$$

$X(T)$ for the spread of agricultural innovations among farmers through personal communications is given by

$$X = \frac{1}{1 + A^{-1}e^{-T}} \quad (11)$$

Recast dynamical equation for the spread of agricultural innovations among farmers through impersonal communications is given by

$$\dot{X} = (X+A)(1-X) \quad (12)$$

$X(T)$ for the spread of agricultural innovations among farmers through impersonal communications is given by

$$X = \frac{1 - e^{-(1+A)T}}{1 + A^{-1}e^{-(1+A)T}} \quad (13)$$

The relative error between the analytical solution and the numerical solution is given by,

$$relativeerror = \frac{analyticalsolution - numericalsolution}{analyticalsolution}$$

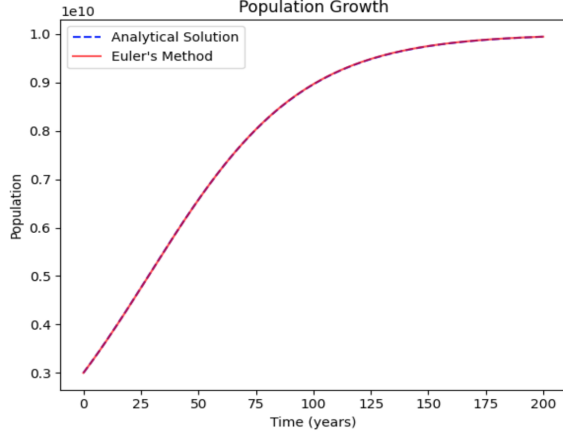
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II. RESULTS

A. Population Growth

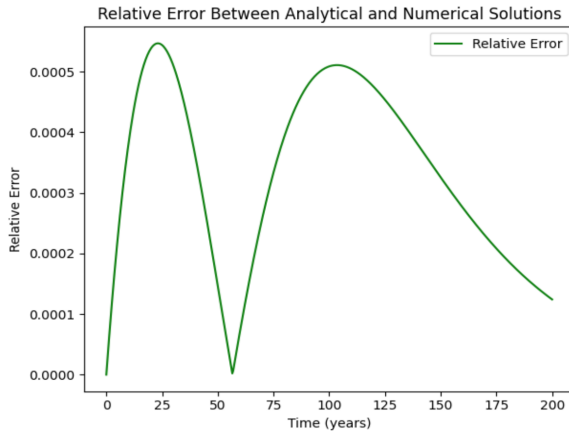
Fig.1 shows the analytical and numerical solutions for $x(t)$. Here $a = 0.03$ per annum and $b = 3 \times 10^{-12}$, based on the global census data of 1961 and $x_0 \approx 3 \times 10^9$ is the initial population.



Estimated limiting population: **10 billion** where the present global population according to the google is **8.2 billion**.

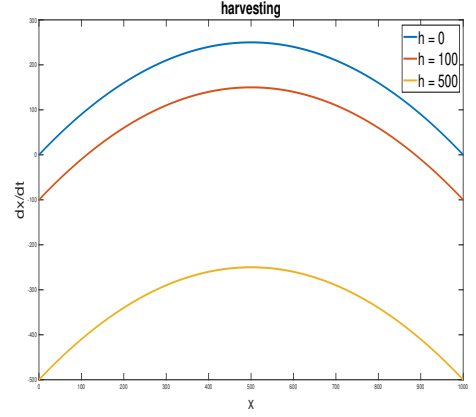
B. Relative Error Analysis

Fig.2 shows the relative error between the analytical and numerical solutions.



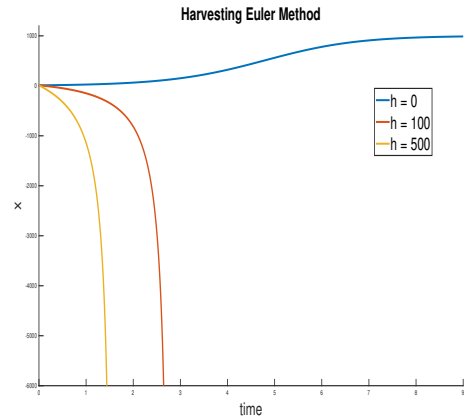
C. Harvesting Model

Fig.3 shows $\dot{x}$ versus x with $h = 0, 100, 500$ with a timestep of 1. Here $r = 1$ and $k = 1000$.



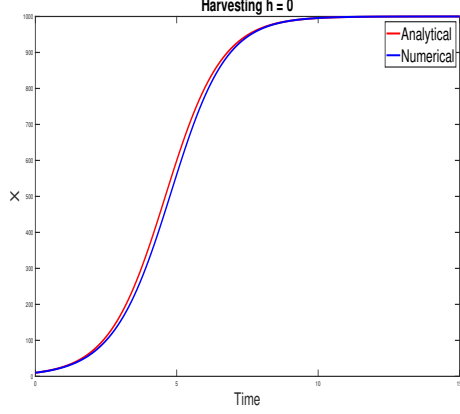
D. Euler's Method

Fig.4 shows x (by Euler's method) versus t with $h = 0, 100, 500$ with a timestep of 0.1. Here the initial value of $x(0) = 10$.



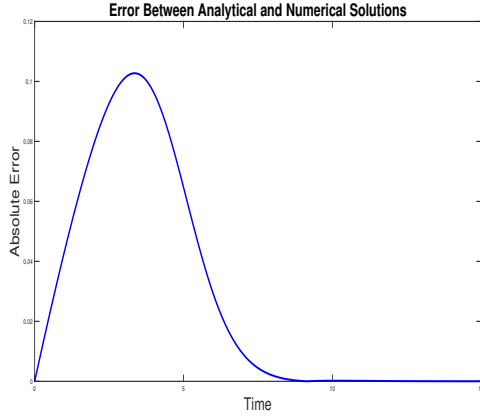
E. Comparison between the analytical and numerical solution

Fig.5 shows comparison between the analytical solution and the numerical solution for $h=0$ with a timestep of 0.1.



F. Relative Error Analysis

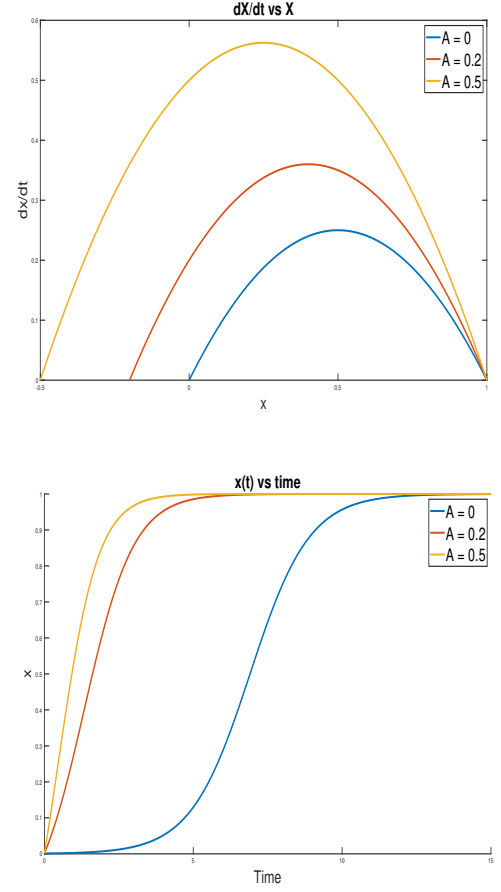
Fig.6 The relative error between the analytical solution and the numerical solution for $h=0$.



G. Spread of agricultural innovations

First figure shows $\dot{X}$ versus X for $A = 0, 0.2, 0.5$ with a timestep of 0.01.

Second figure shows the integral solution $X(T)$ for $A = 0, 0.2, 0.5$ with a timestep of 0.1. Here $X(0)=0.0001$.



III. CONCLUSION

- The logistic population model predicts a limiting population of approximately 10 billion, aligning closely with current global estimates. The numerical solution using Euler's method closely follows the analytical solution with minimal error.
- The spread of agricultural innovations follows a distinct growth patterns depending on communication methods. Personal communication results in logistic-type growth, while impersonal communication accelerates adoption. Increasing A leads to faster saturation.
- The modified logistic equation with harvesting shows that higher harvesting rates (h) significantly impact long-term population sustainability. The numerical approximation using Euler's method closely follows the analytical solution, though relative errors increase over time.